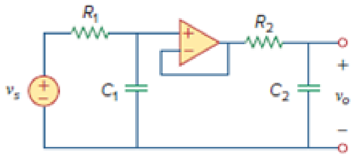
# Chapter 8, Problem 11PP

## Problem

In the op amp circuit shown in Fig. 8.34, $v_s$ = 25$u(t)$ V, find $v_o(t)$ for $t$ > 0. Assume that $R_1$ = $R_2$ = 10 kΩ, $C_1$ = 20 μF, and $C_2$ = 100 μF.

**Figure 8.34:**




## Step-by-step solution

**Step 1** of 14

Refer to Figure 8.34 in the text book for the actual circuit.

At $t<0$, the capacitor acts as open circuit. The value of the voltage source is zero since $u(t)=0$ for $t<0$. That is the voltage source acts as short circuit $t<0$.

Draw the circuit diagram at $t<0$.

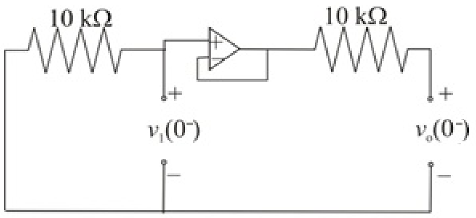



Figure 1

The values of voltages $v_1(0^-)$ and $v_o(0^-)$ are $0\text{ V}$.

**Step 2** of 14

The value of the voltage source is $25\text{ V}$ since $u(t)=1$ for $t\geq 0$.

Draw the circuit diagram at $t>0$.

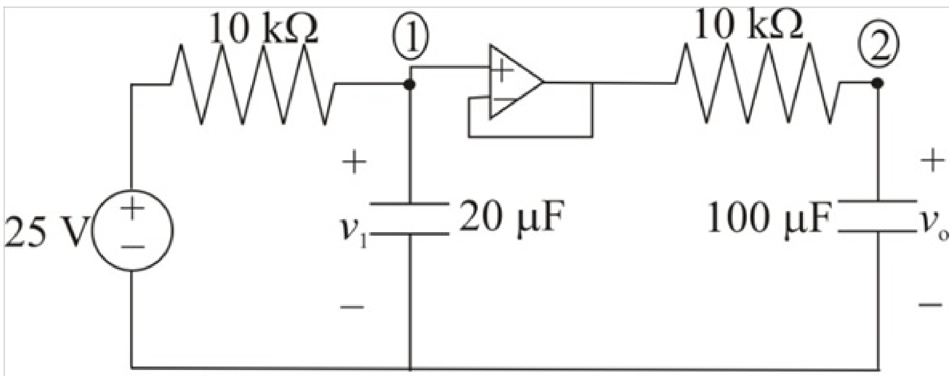



Figure 2

**Step 3** of 14

Buffer amplifier represents same voltage at input and output terminals.

The voltage across the capacitor will not change instantaneously. Hence, at $t>0$ the value of the voltages $v_1(0^+)$ and $v_o(0^+)$ are $0\ \text{V}$.

Notice that the voltage across the capacitor is continuous. Therefore, the value of voltages $v_1(0)$ and $v_o(0)$ at $t=0$ are $0\ \text{V}$.

**Step 4** of 14

Apply Kirchhoff's current law at node-1.

$$\frac{v_1-25}{10\text{ k}}+(20\ \mu)\frac{dv_1}{dt}=0$$

$$v_1-25+0.2\frac{dv_1}{dt}=0 \quad \ldots\ldots (1)$$

Calculate the value of $\frac{dv_1}{dt}$ at $t=0$.

$$0.2\frac{dv_1(0)}{dt}+v_1(0)-25=0$$

Substitute $0\text{ V}$ for $v_1(0)$ and $v_2(0)$.

$$0.2\frac{dv_1(0)}{dt}+(0)-25=0$$

$$0.2\frac{dv_1(0)}{dt}=25$$

$$\frac{dv_1(0)}{dt}=125\text{ V/s}$$

Therefore, the value of the $\frac{dv_1(0)}{dt}$ is $125\text{ V/s}$.

**Step 5** of 14

Short the voltage source and redraw the circuit to estimate the transient behavior of the circuit.

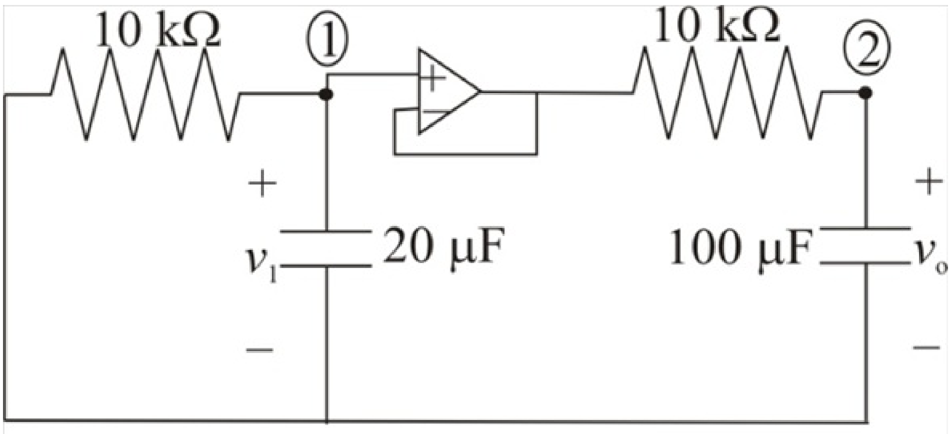


Figure 3

**Step 6** of 14

Apply Kirchhoff's current law at node-1.

$$\frac{v_1}{10\text{ k}}+(20\ \mu)\frac{dv_1}{dt}=0$$

$$v_1+0.2\frac{dv_1}{dt}=0 \quad \ldots\ldots (2)$$

**Step 7** of 14

Apply Kirchhoff's current law at node-2.

$$(100\ \mu)\frac{dv_o}{dt}+\frac{v_o-v_1}{10\text{ k}}=0$$

$$\frac{dv_o}{dt}+v_o-v_1=0$$

$$v_1=\frac{dv_o}{dt}+v_o$$

Substitute $\frac{dv_o}{dt}+v_o$ for $v_1$ in equation (2).

$$\frac{dv_o}{dt}+v_o+0.2\frac{d}{dt}\left(\frac{dv_o}{dt}+v_o\right)=0$$

$$0.2\frac{d^2v_0}{dt^2}+1.2\frac{dv_o}{dt}+v_o=0$$

$$\frac{d^2v_0}{dt^2}+6\frac{dv_o}{dt}+5v_o=0$$

Consider $\frac{di}{dt}$ as $s$.

$$s^2v_0+6sv_0+5v_0=0$$

$$(s^2+6s+5)v_0=0$$

**Step 8** of 14

Consider the characteristic equation of the transient circuit.

$$s^2+6s+5=0$$

$$(s+1)(s+5)=0$$

The roots of the characteristic equation are $s_1=-1$ and $s_2=-5$.

**Step 9** of 14

Consider the expression for the voltage $v_{0t}$ in transient state.

$$v_{0t}=A_1e^{s_1t}+A_2e^{s_2t}$$

Substitute $-1$ for $s_1$ and $-5$ for $s_2$.

$$v_{0t}=A_1e^{-t}+A_2e^{-5t}$$

**Step 10** of 14

At $t = \infty$, as steady state is reached, capacitors are open circuited.

Draw the circuit diagram at $t = \infty$.

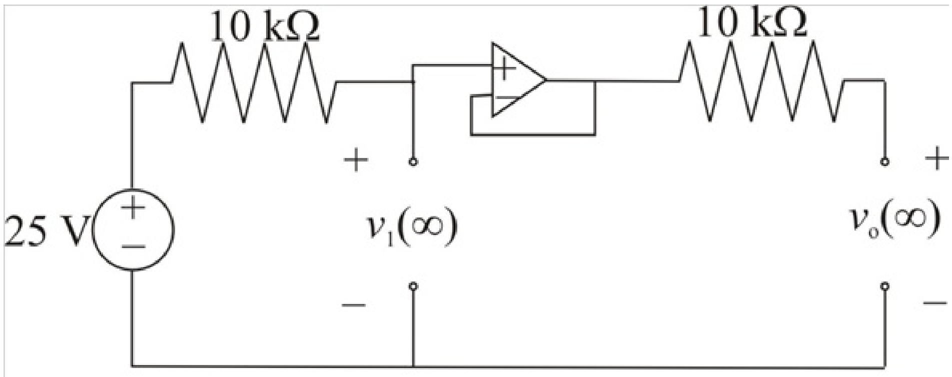



Figure 4

The values of voltages $v_1(\infty)$ and $v_2(\infty)$ are $25 \text{ V}$.

**Step 11** of 14

Notice that the roots are negative and real, hence the circuit exhibits over damped response.

Consider the expression for the voltage $v_1(t)$ for complete response.

$$v_0(t) = v_{0t}(t) + v_{0ss}(t)$$

Here $v_{0t}(t)$ represents the transient response,

$v_{0ss}(t)$ represents the steady state response that is $v_1(\infty)$

$$v_0(t) = v_0 + v_0(\infty)$$

Substitute $25 \text{ V}$ for $v_0(\infty)$ and $A_1e^{-t} + A_2e^{-5t}$ for $v_{0t}$.

$$v_0(t) = A_1e^{-t} + A_2e^{-5t} + 25 \text{ ...... (3)}$$

**Step 12** of 14

Recall equation (3).

$$v_0(t) = A_1e^{-t} + A_2e^{-5t} + 25$$

Substitute $0$ for $t$.

$$v_0(0) = A_1e^{-(0)} + A_2e^{-5(0)} + 25$$

Substitute $0\text{ V}$ for $v_0(0)$.

$$0 = A_1e^{(0)} + A_2e^{(0)} + 25$$
$$0 = A_1(1) + A_2(1) + 25$$
$$0 = A_1 + A_2 + 25$$

$A_1 = -A_2 - 25$ …… (4)

Recall equation (3).

$$v_0(t) = A_1e^{-t} + A_2e^{-5t} + 25$$

Apply derivative with respect to $t$.

$$\begin{aligned}\frac{dv_0(t)}{dt} &= A_1\frac{d\left(e^{-t}\right)}{dt} + A_2\frac{d\left(e^{-5t}\right)}{dt} + \frac{d(25)}{dt} \\ &= -A_1e^{-t} - 5A_2e^{-5t} + 0 \\ &= -A_1e^{-t} - 5A_2e^{-5t}\end{aligned}$$

Substitute $0$ for $t$.

$$\begin{aligned}\frac{dv_0(0)}{dt} &= -A_1e^{(0)} - 5A_2e^{(0)} \\ &= -A_1(1) - 5A_2(1) \\ &= -A_1 - 5A_2\end{aligned}$$

Substitute 0 for $\frac{dv_0(0)}{dt}$ and $-A_2 - 25$ for $A_1$.

$$0 = -(-A_2 - 25) - 5A_2$$
$$0 = A_2 + 25 - 5A_2$$
$$0 - 25 = -4A_2$$

$$\begin{aligned}A_2 &= \frac{25}{4} \\ &= 6.25\end{aligned}$$

Therefore, the value of the constant $A_2$ is $6.25$.

**Step 13** of 14

Recall equation (3).

$$A_1 = -A_2 - 25$$

Substitute 6.25 for $A_2$.

$$\begin{aligned}A_1 &= -(6.25) - 25 \\ &= -31.25\end{aligned}$$

Therefore, the value of the constant $A_1$ is, $-31.25$.

**Step 14** of 14

Recall equation (3).

$$v_0(t) = A_1e^{-t} + A_2e^{-5t} + 25$$

Substitute $6.25$ for $A_2$ and $-31.25$ for $A_1$.

$$v_0(t) = 25 - 31.25e^{-t} + 6.25e^{-5t}$$

Therefore, the value of the voltage $v_0(t)$ is $25 - 31.25e^{-t} + 6.25e^{-5t}$.

Therefore, the value of the voltage $v_0$ is $\boxed{25 - 31.25e^{-t} + 6.25e^{-5t}\ \text{V}, t > 0}$.